SciAR: Scientific Author Representations for scientific authorship identification

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1 Motivation

Authorship verification (AV) asks whether two pieces of text were written by the same person Koppel et al. [2007]. This problem is particularly important in science, where digital libraries must determine which publications belong to which researchers despite ambiguous names, incomplete identifiers such as ORCiD, and inconsistent metadata Rodrigues et al. [2024], OpenAlex [2026]. Accurate author identities also support downstream systems such as literature search, bibliometrics, reviewer recommendation, and conflict-of-interest detection Mihaljević and Santamaría [2021], Andrade-Navarro et al. [2012].

Large-scale author name disambiguation (AND) systems combine many signals to determine whether publication records belong to the same researcher, including name similarity, coauthors, affiliations, citations, persistent identifiers, venues, and research topics OpenAlex [2026], Subramanian et al. [2024]. However, many of these signals depend on bibliographic metadata that may be missing, incomplete, or unreliable. When richer identity signals are unavailable, the semantic content or topic of a researcher's publications remains the main signal for determining which works should be grouped together. Stylometric information, or characteristics of "how" an author writes rather than "what" they write about, is largely absent from the feature sets used by major scientific AND systems. Although scientific writing is highly standardized, previous work has demonstrated that scientific articles contain measurable author-specific stylistic variation Bergsma et al. [2012]. We propose learning scientific writing style as a complementary signal for scientific authorship while explicitly controlling for topic, training the model to distinguish authors whose work is semantically similar.

A reliable representation of scientific writing style could support several downstream applications. Most directly, it could provide an additional signal for authorship verification and author name disambiguation when bibliographic metadata is sparse or ambiguous. It could also enable studies of how an individual's scientific writing style changes over the course of a career, across collaborations, or between research areas.

Our input space will consist of pairs of scientific texts, initially titles and abstracts and potentially full text. The output will be a probability or similarity score representing whether the two texts have the same first author. We will construct a large dataset from a scholarly corpus such as OpenAlex, S2ORC, or NASA ADS, using ORCID-associated records where possible to establish high-confidence author identities. In order to prevent topical similarity from being a signal the models latch onto, we will enforce that our training pairs have a high topical similarity according to a topical scientific embedding model such as SPECTER2 that way topic does not make any two works in a pair separable.

Language-model-based authorship systems already exist, but none for scientific work. LUAR learns general-purpose authorship representations, while CISR specifically attempts to separate writing style from content by controlling topic during authorship-verification training. We will evaluate models such as these alongside BERT and scientific-domain models such as SciBERT, potentially fine-tuned using contrastive or metric-learning objectives.

In the context of increasingly common language-model-assisted writing, stylometric representations may also help study how human author signals change when scientific text is edited, paraphrased, or generated by language models. Learning representations of human scientific writing style may eventually help study how identifiable human authorship signals persist, disappear, or change when language models generate or rewrite scientific work.

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